Architecture Refactor Report

convertfloww · FlowwSuite

5 HIGH | 4 MEDIUM | 5 LOW | 7 STEPS

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150 / 150 tests passing · TypeScript strict · ESLint clean

Ø=Ý4 High-Priority Issues (H1 – H5)

H1 God component — 412 lines, 3 responsibilities

shared/components/SettingsMenu.tsx

Issue: Mixes Accessibility/Theme settings, Profile CRUD (name/password/avatar), and Danger Zone delete into a single file. createProfileRepository() is instantiated 3× inside separate click handlers.

Fix: Extract AccessibilitySettings, ThemeSettings, ProfileSettings, and DangerZone panel components. Add a useSettingsActions() hook with a stable repo ref. SettingsMenu becomes a ~60-line shell.

H2 Monolith view — 611 lines, 4 concerns inlined

attendance/components/ProfileSelection.tsx

Issue: Profile grid, Add-profile modal, Password-unlock prompt with lockout logic, and avatar picker are all in one file. Business rules (lockout after 5 attempts, bcrypt migration) sit alongside JSX.

Fix: Extract AddProfileModal, PasswordUnlockModal, and ProfileCard components. Move AVATARS constant to attendance/constants.ts. ProfileSelection becomes a ~150-line coordinator.

H3 Raw SQL in hook — no data layer

admin/hooks/useAdminDashboard.ts

Issue: Seven separate supabase.from() calls spread across the hook file. moderateAll is defined inline inside the return object. AdminProfile and Announcement types duplicate concepts from other features.

Fix: Create AdminRepository class encapsulating all DB access. useAdminDashboard becomes coordination logic only (state + toast). Hook drops from 238 to ~120 lines.

H4 Raw Supabase query inside render component

attendance/components/AttendanceView.tsx

Issue: Announcement fetch + mark_announcement_seen RPC are in a useEffect directly in the view (lines 86–117). Announcement interface re-declared here; it already exists in admin.types.ts.

Fix: Create useAnnouncement(profileId) hook returning { announcement, dismiss }. Import Announcement type from admin.types.ts. AttendanceView calls the hook instead of containing raw fetch logic.

H5

DOM side-effects at module load time

shared/store/useThemeStore.ts

Issue: document.documentElement.classList.add runs during store initialisation (line 20–22), before React mounts. Breaks test environments and SSR; causes flash-of-wrong-theme in some scenarios.

Fix: Move DOM mutations (classList.add / setAttribute) into the Zustand persist onRehydrateStorage callback. Store becomes pure; tests no longer need document mocking.

Ø=ßá Medium-Priority Issues (M1 – M4)

M1 Async side-effects and self-reference inside store

attendance/store/useAttendanceStore.ts

Issue: uploadAvatar and syncFromCloud call useAttendanceStore.getState() from within a store action — a self-reference anti-pattern that prevents testing actions in isolation.

Fix: Move uploadAvatar and syncFromCloud into a useAttendanceSync() hook that reads store state externally via selectors.

M2 Repository instantiated 3× inside event handlers

shared/components/SettingsMenu.tsx

Issue: createProfileRepository(supabase) is called fresh on every button click instead of once at component mount level.

Fix: Resolved by Step 1 (useSettingsActions hook holds a stable repo ref).

M3 Module-level mutable globals for render queue

pdf-engine/hooks/usePdfDocument.ts

Issue: renderQueue[] and isQueueRunning are module-level variables that persist between Vitest test cases and across HMR reloads.

Fix: Encapsulate in a RenderQueue class; instantiate via useMemo inside the hook.

M4 Auth listener registered at module import

core/lib/useAuthStore.ts

Issue: supabase.auth.onAuthStateChange is called at line 66 when the module loads, before React mounts. If the store is imported before Supabase initialises, the listener is silently dropped.

Fix: Export an initAuth() function; call it in main.tsx after createRoot().render().

Ø=ßâ Low-Priority Issues (L1 – L5)

L1 window.location.href for SPA navigation

src/App.tsx

Fix: Use useNavigate() from react-router-dom.

L2

Announcement type re-declared

attendance/components/AttendanceView.tsx

Fix: Import Announcement from admin.types.ts; remove local re-declaration.

L3

Magic avatar URL array inlined in component

attendance/components/ProfileSelection.tsx

Fix: Move to src/features/attendance/constants.ts.

L4

moderateAll defined inline in return object

admin/hooks/useAdminDashboard.ts

Fix: Extract to a named const moderateAll = async () => { ... } before the return.

L5

Vestigial empty directory

src/store/

Fix: Delete the directory.

Ø=ÜË 7-Step Refactor Roadmap

#	Title	Files Affected	Fixes	Effort
1	Split SettingsMenu into sub-components	SettingsMenu.tsx !' AccessibilitySetti ProfileSettings, DangerZone + useSettingsActions hook	**H1, M2**	Medium
2	Extract useAnnouncement hook	New: attendance/hooks/useAnnouncement.ts	**H4, L2**	Small
3	Create AdminRepository	New: admin/repository/adminRepository.ts; refactor useAdminDashboard.ts	**H3, L4**	Medium
4	Split ProfileSelection into sub-components	New: AddProfileModal, PasswordUnlockModal, ProfileCard; attendance/constants.ts	**H2, L3**	Large
5	Fix store side-effects	useThemeStore.ts, useAuthStore.ts, main.tsx	**H5, M4**	Small
6	Scope RenderQueue in usePdfDocument	pdf-engine/hooks/usePdfDocument.ts	**M3**	Small
7	Fix SPA navigation + remove /store	App.tsx, delete src/store/	**L1, L5**	Trivial

Ø=ÝÂ Current Structure

```
src/
    core/lib/              !• supabase client, useAuthStore
  features/
      admin/               !• AdminLayout, useAdminDashboard
      attendance/          !• AttendanceView, ProfileSelection
      pdf-engine/          !• PdfEngineView, usePdfDocument
  shared/
      components/          !• SettingsMenu (412 ln), BottomTabs
      store/               !• useThemeStore
    store/                 !• EMPTY — vestigial
```